

FIGURE 9.14.7 Curve  $C$  for Problem 6

7.  $\mathbf{F} = xy\mathbf{i} + 2yz\mathbf{j} + xz\mathbf{k}$ ;  $C$  the boundary given in Problem 6
8.  $\mathbf{F} = (x + 2z)\mathbf{i} + (3x + y)\mathbf{j} + (2y - z)\mathbf{k}$ ;  $C$  the curve of intersection of the plane  $x + 2y + z = 4$  with the coordinate planes
9.  $\mathbf{F} = y^3\mathbf{i} - x^3\mathbf{j} + z^3\mathbf{k}$ ;  $C$  the trace of the cylinder  $x^2 + y^2 = 1$  in the plane  $x + y + z = 1$
10.  $\mathbf{F} = x^2y\mathbf{i} + (x + y^2)\mathbf{j} + xy^2z\mathbf{k}$ ;  $C$  the boundary of the surface shown in FIGURE 9.14.8

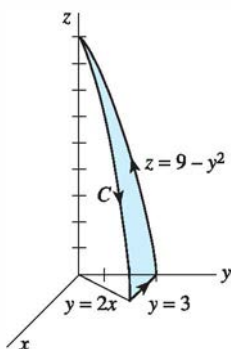


FIGURE 9.14.8 Curve  $C$  for Problem 10

11.  $\mathbf{F} = x\mathbf{i} + x^3y^2\mathbf{j} + z\mathbf{k}$ ;  $C$  the boundary of the semi-ellipsoid  $z = \sqrt{4 - 4x^2 - y^2}$  in the plane  $z = 0$
12.  $\mathbf{F} = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$ ;  $C$  the curve of intersection of the plane  $x + y + z = 0$  and the sphere  $x^2 + y^2 + z^2 = 1$  [Hint: Recall that the area of an ellipse  $x^2/a^2 + y^2/b^2 = 1$  is  $\pi ab$ .]

In Problems 13–16, use Stokes' theorem to evaluate  $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS$ . Assume that the surface  $S$  is oriented upward.

13.  $\mathbf{F} = 6yz\mathbf{i} + 5x\mathbf{j} + yze^{x^2}\mathbf{k}$ ;  $S$  that portion of the paraboloid  $z = \frac{1}{4}x^2 + y^2$  for  $0 \leq z \leq 4$
14.  $\mathbf{F} = y\mathbf{i} + (y - x)\mathbf{j} + z^2\mathbf{k}$ ;  $S$  that portion of the sphere  $x^2 + y^2 + (z - 4)^2 = 25$  for  $z \geq 0$
15.  $\mathbf{F} = 3x^2\mathbf{i} + 8x^3y\mathbf{j} + 3x^2y\mathbf{k}$ ;  $S$  that portion of the plane  $z = x$  that lies inside the rectangular cylinder defined by the planes  $x = 0, y = 0, x = 2, y = 2$
16.  $\mathbf{F} = 2xy^2z\mathbf{i} + 2x^2yz\mathbf{j} + (x^2y^2 - 6x)\mathbf{k}$ ;  $S$  that portion of the plane  $z = y$  that lies inside the cylinder  $x^2 + y^2 = 1$
17. Use Stokes' theorem to evaluate

$$\oint_C z^2 e^{x^2} dx + xy^2 dy + \tan^{-1} y \, dz$$

where  $C$  is the circle  $x^2 + y^2 = 9$ , by finding a surface  $S$  with  $C$  as its boundary and such that the orientation of  $C$  is counterclockwise as viewed from above.

18. Consider the surface integral  $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS$ , where  $\mathbf{F} = xyz\mathbf{k}$  and  $S$  is that portion of the paraboloid  $z = 1 - x^2 - y^2$  for  $z \geq 0$  oriented upward.
  - (a) Evaluate the surface integral by the method of Section 9.13; that is, do not use Stokes' theorem.
  - (b) Evaluate the surface integral by finding a simpler surface that is oriented upward and has the same boundary as the paraboloid.
  - (c) Use Stokes' theorem to verify the result in part (b).

## 9.15 Triple Integrals

**Introduction** The steps leading to the definition of the *three-dimensional definite integral* or **triple integral** are quite similar to the steps leading to the definition of the double integral. Obvious differences: instead of a function of two variables we are integrating a function  $f$  of three variables, not over a region  $R$  in a coordinate plane, but over a region  $D$  of 3-space.

1. Let  $w = F(x, y, z)$  be defined over a closed and bounded region  $D$  of space.
2. By means of a three-dimensional grid of vertical and horizontal planes parallel to the coordinate planes, form a partition  $P$  of  $D$  into  $n$  subregions (boxes)  $D_k$  of volumes  $\Delta V_k$  that lie entirely in  $D$ .
3. Let  $\|P\|$  be the **norm** of the partition or the length of the longest diagonal of the  $D_k$ .
4. Choose a sample point  $(x_k^*, y_k^*, z_k^*)$  in each subregion  $D_k$ . See FIGURE 9.15.1.
5. Form the sum  $\sum_{k=1}^n F(x_k^*, y_k^*, z_k^*) \Delta V_k$ .

A sum of the form  $\sum_{k=1}^n F(x_k^*, y_k^*, z_k^*) \Delta V_k$ , where  $(x_k^*, y_k^*, z_k^*)$  is an arbitrary point within each  $D_k$  and  $\Delta V_k$  denotes the volume of each  $D_k$ , is called a **Riemann sum**. The type of partition used in step 2, where all the  $D_k$  lie completely within  $D$ , is called an **inner partition** of  $D$ .

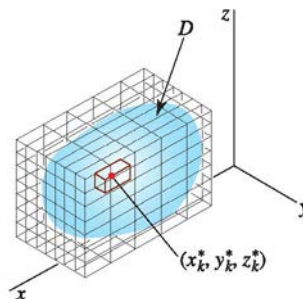


FIGURE 9.15.1 Sample point in  $k$ th subregion

**Definition 9.15.1 The Triple Integral**

Let  $F$  be a function of three variables defined over a closed region  $D$  of 3-space. Then the **triple integral of  $F$  over  $D$**  is given by

$$\iiint_D F(x, y, z) dV = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n F(x_k^*, y_k^*, z_k^*) \Delta V_k. \quad (1)$$

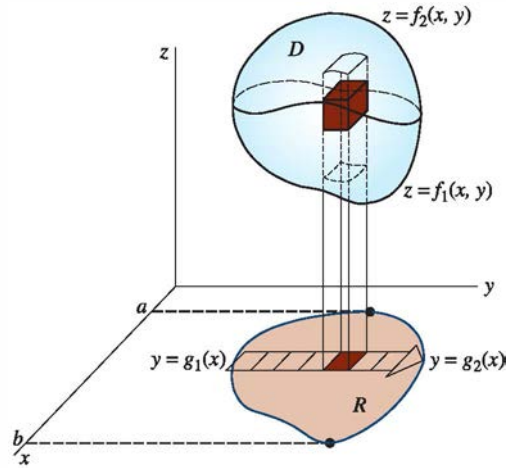
As in our previous discussions on the integral, when  $F$  is continuous over  $D$ , the limit in (1) exists; that is,  $F$  is **integrable** over  $D$ .

**Evaluation by Iterated Integrals** If the region  $D$  is bounded above by the graph of  $z = f_2(x, y)$  and bounded below by the graph of  $z = f_1(x, y)$ , then it can be shown that the triple integral (1) can be expressed as a double integral of the partial integral  $\int_{f_1(x, y)}^{f_2(x, y)} F(x, y, z) dz$ ; that is,

$$\iiint_D F(x, y, z) dV = \iint_R \left[ \int_{f_1(x, y)}^{f_2(x, y)} F(x, y, z) dz \right] dA,$$

where  $R$  is the orthogonal projection of  $D$  onto the  $xy$ -plane. In particular, if  $R$  is a Type I region, then, as shown in **FIGURE 9.15.2**, the triple integral of  $F$  over  $D$  can be written as an iterated integral:

$$\iiint_D F(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{f_1(x, y)}^{f_2(x, y)} F(x, y, z) dz dy dx. \quad (2)$$



**FIGURE 9.15.2** Geometric interpretation of (2)

To evaluate the iterated integral in (2) we begin by evaluating the partial integral

$$\int_{f_1(x, y)}^{f_2(x, y)} F(x, y, z) dz,$$

in which *both*  $x$  and  $y$  are held fixed.

In a double integral there are only two possible orders of integration:  $dy dx$  and  $dx dy$ . The triple integral in (2) illustrates one of *six* possible orders of integration:

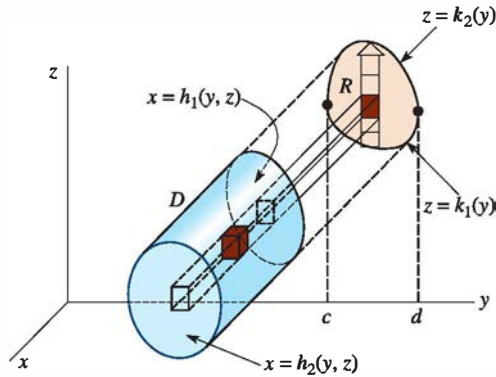
$$\begin{aligned} dz dy dx, & \quad dz dx dy, & \quad dy dx dz, \\ dx dy dz, & \quad dx dz dy, & \quad dy dz dx. \end{aligned}$$

The last two differentials tell the coordinate plane in which the region  $R$  is situated. For example, the iterated integral corresponding to the order of integration  $dx dz dy$  must have

the form

$$\iiint_D F(x, y, z) \, dV = \int_c^d \int_{k_1(y)}^{k_2(y)} \int_{h_1(y, z)}^{h_2(y, z)} F(x, y, z) \, dx \, dz \, dy.$$

The geometric interpretation of this integral and the region  $R$  of integration in the  $yz$ -plane are shown in **FIGURE 9.15.3**.



**FIGURE 9.15.3** Integration: first  $x$ , then  $z$ , then  $y$

**Applications** A list of some of the standard applications of the triple integral follows:

**Volume:** If  $F(x, y, z) = 1$ , then the **volume** of the solid  $D$  is

$$V = \iiint_D dV.$$

**Mass:** If  $\rho(x, y, z)$  is density, then the **mass** of the solid  $D$  is given by

$$m = \iiint_D \rho(x, y, z) \, dV.$$

**First Moments:** The **first moments** of the solid about the coordinate planes indicated by the subscripts are given by

$$M_{xy} = \iiint_D z\rho(x, y, z) \, dV, \quad M_{xz} = \iiint_D y\rho(x, y, z) \, dV,$$

$$M_{yz} = \iiint_D x\rho(x, y, z) \, dV.$$

**Center of Mass:** The coordinates of the **center of mass** of  $D$  are given by

$$\bar{x} = \frac{M_{yz}}{m}, \quad \bar{y} = \frac{M_{xz}}{m}, \quad \bar{z} = \frac{M_{xy}}{m}.$$

**Centroid:** If  $\rho(x, y, z) = \text{a constant}$ , the center of mass is called the **centroid** of the solid.

**Second Moments:** The **second moments**, or **moments of inertia** of  $D$  about the coordinate axes indicated by the subscripts, are given by

$$I_x = \iiint_D (y^2 + z^2)\rho(x, y, z) \, dV, \quad I_y = \iiint_D (x^2 + z^2)\rho(x, y, z) \, dV, \quad I_z = \iiint_D (x^2 + y^2)\rho(x, y, z) \, dV.$$

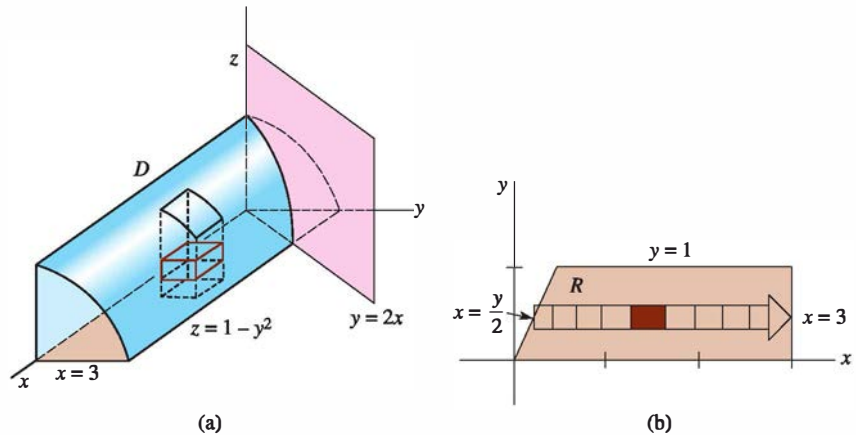
**Radius of Gyration:** As in Section 9.10, if  $I$  is a moment of inertia of the solid about a given axis, then the **radius of gyration** is

$$R_g = \sqrt{\frac{I}{m}}.$$

### EXAMPLE 1 Volume of a Solid

Find the volume of the solid in the first octant bounded by the graphs of  $z = 1 - y^2$ ,  $y = 2x$ , and  $x = 3$ .

**SOLUTION** As indicated in **FIGURE 9.15.4(a)**, the first integration with respect to  $z$  is from 0 to  $1 - y^2$ . Furthermore, from Figure 9.15.4(b) we see that the projection of the solid  $D$  in the



**FIGURE 9.15.4** Solid  $D$  and region  $R$  of integration in Example 1

$xy$ -plane is a region of Type II. Hence, we next integrate, with respect to  $x$ , from  $y/2$  to 3. The last integration is with respect to  $y$  from 0 to 1. Thus,

$$\begin{aligned} V &= \iiint_D dV = \int_0^1 \int_{y/2}^3 \int_0^{1-y^2} dz \, dx \, dy \\ &= \int_0^1 \int_{y/2}^3 (1 - y^2) \, dx \, dy \\ &= \int_0^1 \left[ x - xy^2 \right]_{y/2}^3 dy \\ &= \int_0^1 \left( 3 - 3y^2 - \frac{1}{2}y + \frac{1}{2}y^3 \right) dy \\ &= \left[ 3y - y^3 - \frac{1}{4}y^2 + \frac{1}{8}y^4 \right]_0^1 = \frac{15}{8}. \end{aligned}$$

### EXAMPLE 2 Changing the Order of Integration

Change the order of integration in

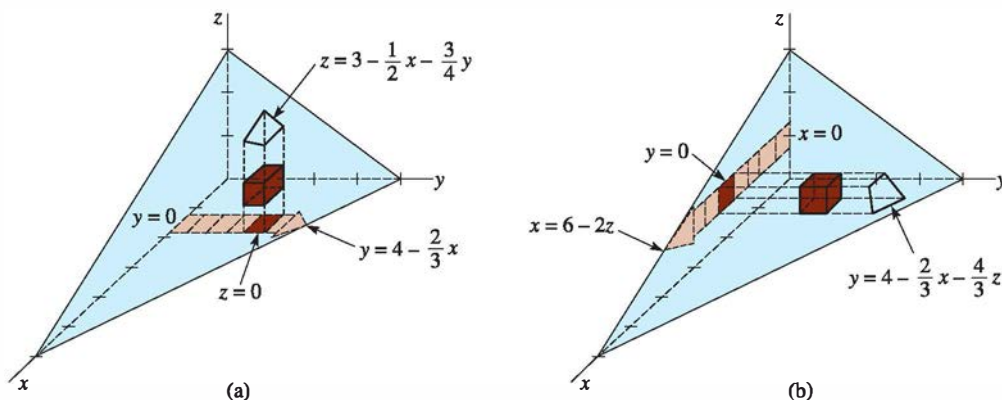
$$\int_0^6 \int_0^{4-2x/3} \int_0^{3-x/2-3y/4} F(x, y, z) \, dz \, dy \, dx$$

to  $dy \, dx \, dz$ .

**SOLUTION** As seen in **FIGURE 9.15.5(a)**, the region  $D$  is the solid in the first octant bounded by the three coordinate planes and the plane  $2x + 3y + 4z = 12$ . Referring to Figure 9.15.5(b) and the table, we conclude that

$$\int_0^6 \int_0^{4-2x/3} \int_0^{3-x/2-3y/4} F(x, y, z) \, dz \, dy \, dx = \int_0^3 \int_0^{6-2z} \int_0^{4-2x/3-4z/3} F(x, y, z) \, dy \, dx \, dz.$$

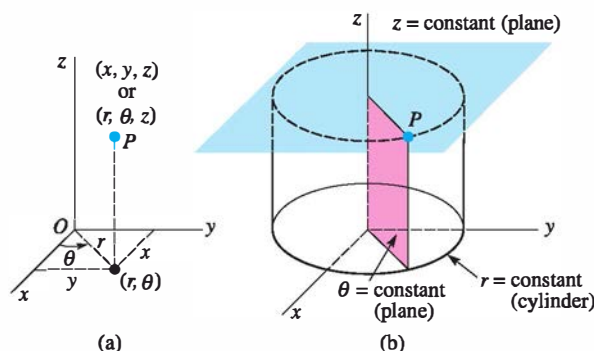
| Order of Integration | First Integration      | Second Integration | Third Integration |
|----------------------|------------------------|--------------------|-------------------|
| $dz \, dy \, dx$     | 0 to $3 - x/2 - 3y/4$  | 0 to $4 - 2x/3$    | 0 to 6            |
| $dy \, dx \, dz$     | 0 to $4 - 2x/3 - 4z/3$ | 0 to $6 - 2z$      | 0 to 3            |



**FIGURE 9.15.5** Changing order of integration in Example 2

Depending on the geometry of a region in 3-space, the evaluation of a triple integral over that region may be made easier by utilizing a new coordinate system.

**Cylindrical Coordinates** The cylindrical coordinate system combines the polar description of a point in the plane with the rectangular description of the  $z$ -component of a point in space. As seen in **FIGURE 9.15.6(a)**, the cylindrical coordinates of a point  $P$  are denoted by the ordered triple  $(r, \theta, z)$ . The word *cylindrical* arises from the fact that a point  $P$  in space is determined by the intersection of the planes  $z = \text{constant}$  and  $\theta = \text{constant}$  with a cylinder  $r = \text{constant}$ . See Figure 9.15.6(b).



**FIGURE 9.15.6** Cylindrical coordinates

**Conversion of Cylindrical Coordinates to Rectangular Coordinates** From Figure 9.15.6(a) we also see that the rectangular coordinates  $(x, y, z)$  of a point can be obtained from the cylindrical coordinates  $(r, \theta, z)$  by means of

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z. \quad (3)$$

### EXAMPLE 3 Cylindrical to Rectangular Coordinates

Convert  $(8, \pi/3, 7)$  in cylindrical coordinates to rectangular coordinates.

**SOLUTION** From (3),

$$x = 8 \cos \frac{\pi}{3} = 4, \quad y = 8 \sin \frac{\pi}{3} = 4\sqrt{3}, \quad z = 7.$$

Thus,  $(8, \pi/3, 7)$  is equivalent to  $(4, 4\sqrt{3}, 7)$  in rectangular coordinates.

**Conversion of Rectangular Coordinates to Cylindrical Coordinates** To express rectangular coordinates  $(x, y, z)$  as cylindrical coordinates, we use

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}, \quad z = z. \quad (4)$$

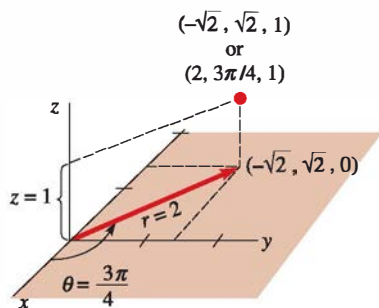


FIGURE 9.15.7 Converting rectangular to cylindrical coordinates in Example 4

#### EXAMPLE 4 Rectangular to Cylindrical Coordinates

Convert  $(-\sqrt{2}, \sqrt{2}, 1)$  in rectangular coordinates to cylindrical coordinates.

**SOLUTION** From (4) we see that

$$r^2 = (-\sqrt{2})^2 + (\sqrt{2})^2 = 4, \quad \tan \theta = \frac{\sqrt{2}}{-\sqrt{2}} = -1, \quad z = 1.$$

If we take  $r = 2$ , then, consistent with the fact that  $x < 0$  and  $y > 0$ , we take  $\theta = 3\pi/4$ .<sup>\*</sup> Consequently,  $(-\sqrt{2}, \sqrt{2}, 1)$  is equivalent to  $(2, 3\pi/4, 1)$  in cylindrical coordinates. See FIGURE 9.15.7.

**Triple Integrals in Cylindrical Coordinates** Recall from Section 9.11 that the area of a polar rectangle is  $\Delta A = r^* \Delta r \Delta \theta$ , where  $r^*$  is the average radius. From FIGURE 9.15.8(a) we see that the volume of a cylindrical wedge is simply  $\Delta V = (\text{area of base})(\text{height}) = r^* \Delta r \Delta \theta \Delta z$ . Thus, if  $F(r, \theta, z)$  is a continuous function over the region  $D$ , as shown in Figure 9.15.8(b), then the triple integral of  $F$  over  $D$  is given by

$$\iiint_D F(r, \theta, z) dV = \iint_R \left[ \int_{f_1(r, \theta)}^{f_2(r, \theta)} F(r, \theta, z) dz \right] dA = \int_{\alpha}^{\beta} \int_{g_1(\theta)}^{g_2(\theta)} \int_{f_1(r, \theta)}^{f_2(r, \theta)} F(r, \theta, z) r dz dr d\theta.$$

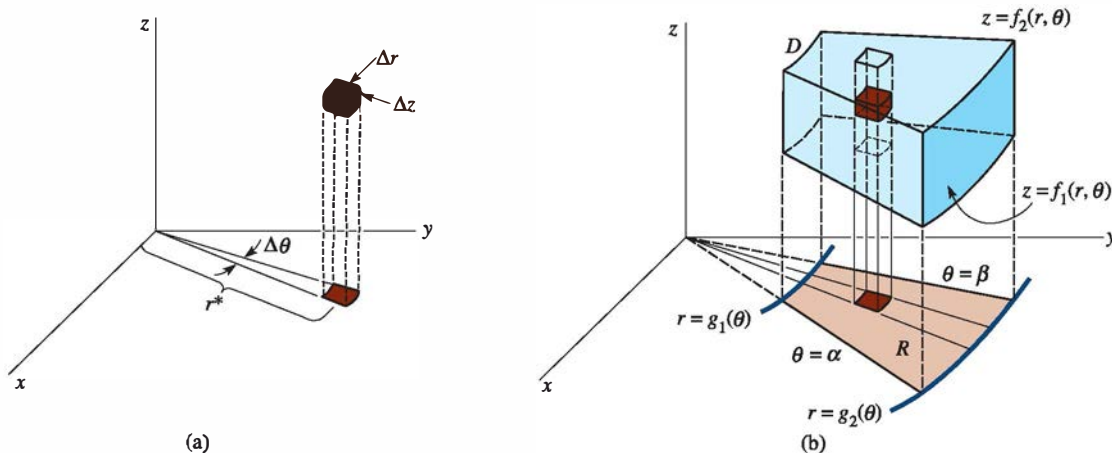


FIGURE 9.15.8 Cylindrical wedge in (a); region  $D$  in (b)

#### EXAMPLE 5 Center of Mass

A solid in the first octant has the shape determined by the graph of the cone  $z = \sqrt{x^2 + y^2}$  and the planes  $z = 1$ ,  $x = 0$ , and  $y = 0$ . Find the center of mass if the density is given by  $\rho(r, \theta, z) = r$ .

**SOLUTION** In view of (4), the equation of the cone is  $z = r$ . Hence, we see from FIGURE 9.15.9 that

$$\begin{aligned} m &= \iiint_D r dV = \int_0^{\pi/2} \int_0^1 \int_r^1 r(r dz dr d\theta) \\ &= \int_0^{\pi/2} \int_0^1 r^2 z \Big|_r^1 dr d\theta \\ &= \int_0^{\pi/2} \int_0^1 (r^2 - r^3) dr d\theta = \frac{\pi}{24} \end{aligned}$$

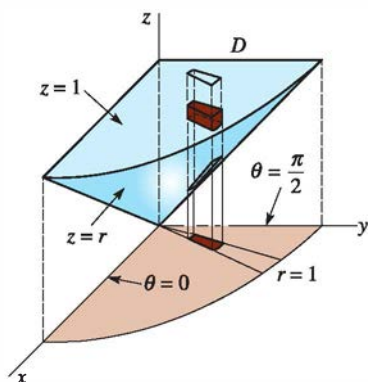


FIGURE 9.15.9 Solid in Example 5

<sup>\*</sup>If we use  $\theta = \tan^{-1}(-1) = -\pi/4$ , then we can use  $r = -2$ . Notice that the combinations  $r = 2$ ,  $\theta = -\pi/4$  and  $r = -2$ ,  $\theta = 3\pi/4$  are inconsistent.



$$\begin{aligned}
 M_{xy} &= \iiint_D z r \, dV = \int_0^{\pi/2} \int_0^1 \int_r^1 z r^2 \, dz \, dr \, d\theta \\
 &= \int_0^{\pi/2} \int_0^1 \left[ \frac{z^2}{2} r^2 \right]_r^1 \, dr \, d\theta \\
 &= \frac{1}{2} \int_0^{\pi/2} \int_0^1 (r^2 - r^4) \, dr \, d\theta = \frac{\pi}{30}.
 \end{aligned}$$

In the integrals for  $M_{xz}$  and  $M_{yz}$  we substitute  $y = r \sin \theta$  and  $x = r \cos \theta$ :

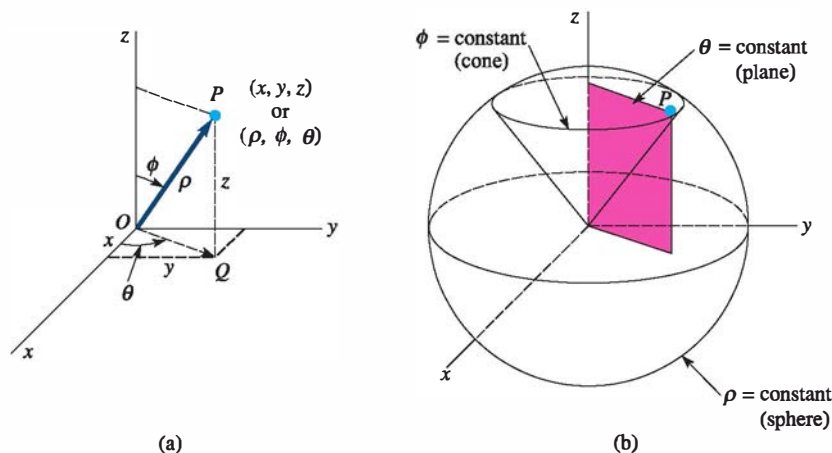
$$\begin{aligned}
 M_{xz} &= \iiint_D r^2 \sin \theta \, dV = \int_0^{\pi/2} \int_0^1 \int_r^1 r^3 \sin \theta \, dz \, dr \, d\theta \\
 &= \int_0^{\pi/2} \int_0^1 \left[ r^3 z \sin \theta \right]_r^1 \, dr \, d\theta \\
 &= \int_0^{\pi/2} \int_0^1 (r^3 - r^4) \sin \theta \, dr \, d\theta = \frac{1}{20} \\
 M_{yz} &= \iiint_D r^2 \cos \theta \, dV = \int_0^{\pi/2} \int_0^1 \int_r^1 r^3 \cos \theta \, dz \, dr \, d\theta = \frac{1}{20}.
 \end{aligned}$$

Hence,

$$\bar{x} = \frac{M_{yz}}{m} = \frac{1/20}{\pi/24} \approx 0.38, \quad \bar{y} = \frac{M_{xz}}{m} = \frac{1/20}{\pi/24} \approx 0.38, \quad \bar{z} = \frac{M_{xy}}{m} = \frac{\pi/30}{\pi/24} \approx 0.8.$$

The center of mass has the approximate coordinates **(0.38, 0.38, 0.8)**.  $\equiv$

**Spherical Coordinates** As seen in **FIGURE 9.15.10(a)**, the **spherical coordinates** of a point  $P$  are given by the ordered triple  $(\rho, \phi, \theta)$ , where  $\rho$  is the distance from the origin to  $P$ ,  $\phi$  is the angle between the positive  $z$ -axis and the vector  $\overrightarrow{OP}$ , and  $\theta$  is the angle measured from the positive  $x$ -axis to the vector projection  $\overrightarrow{OQ}$  of  $\overrightarrow{OP}$ . \* Figure 9.15.10(b) shows that a point  $P$  in space is determined by the intersection of a cone  $\phi = \text{constant}$ , a plane  $\theta = \text{constant}$ , and a sphere  $\rho = \text{constant}$ ; whence arises the name “spherical” coordinates.



**FIGURE 9.15.10** Spherical coordinates

**Conversion of Spherical Coordinates to Rectangular and Cylindrical Coordinates** To transform from spherical coordinates  $(\rho, \phi, \theta)$  to rectangular coordinates  $(x, y, z)$ , we observe from Figure 9.15.10(a) that

$$x = \|\overrightarrow{OQ}\| \cos \theta, \quad y = \|\overrightarrow{OQ}\| \sin \theta, \quad z = \|\overrightarrow{OP}\| \cos \phi.$$

\* $\theta$  is the same angle as in polar and cylindrical coordinates.

Since  $|\overrightarrow{OQ}| = \rho \sin \phi$  and  $|\overrightarrow{OP}| = \rho$ , the foregoing equations become

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi. \quad (5)$$

It is customary to take  $\rho \geq 0$  and  $0 \leq \phi \leq \pi$ . Also, since  $|\overrightarrow{OQ}| = \rho \sin \phi = r$ , the formulas

$$r = \rho \sin \phi, \quad \theta = \theta, \quad z = \rho \cos \phi, \quad (6)$$

enable us to transform from spherical coordinates  $(\rho, \phi, \theta)$  to cylindrical coordinates  $(r, \theta, z)$ .

### EXAMPLE 6 Spherical to Rectangular and Cylindrical Coordinates

Convert  $(6, \pi/4, \pi/3)$  in spherical coordinates to rectangular coordinates and cylindrical coordinates.

**SOLUTION** Identifying  $\rho = 6$ ,  $\phi = \pi/4$ , and  $\theta = \pi/3$ , we find from (5) that

$$x = 6 \sin \frac{\pi}{4} \cos \frac{\pi}{3} = \frac{3\sqrt{2}}{2}, \quad y = 6 \sin \frac{\pi}{4} \sin \frac{\pi}{3} = \frac{3\sqrt{6}}{2}, \quad z = 6 \cos \frac{\pi}{4} = 3\sqrt{2}.$$

The rectangular coordinates of the point are  $(3\sqrt{2}/2, 3\sqrt{6}/2, 3\sqrt{2})$ .

From (6) we obtain

$$r = 6 \sin \frac{\pi}{4} = 3\sqrt{2}, \quad \theta = \frac{\pi}{3}, \quad z = 6 \cos \frac{\pi}{4} = 3\sqrt{2}.$$

Thus, the cylindrical coordinates of the point are  $(3\sqrt{2}, \pi/3, 3\sqrt{2})$ .  $\equiv$

**Conversion of Rectangular Coordinates to Spherical Coordinates** To transform from rectangular coordinates to spherical coordinates, we use

$$\rho^2 = x^2 + y^2 + z^2, \quad \tan \theta = \frac{y}{x}, \quad \cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}. \quad (7)$$

**Triple Integrals in Spherical Coordinates** As seen in **FIGURE 9.15.11**, the volume of a **spherical wedge** is given by the approximation

$$\Delta V \approx \rho^2 \sin \phi \Delta \rho \Delta \phi \Delta \theta.$$

Thus, in a triple integral of a continuous spherical coordinate function  $F(\rho, \phi, \theta)$ , the differential of volume  $dV$  is given by

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

A typical triple integral in spherical coordinates has the form

$$\iiint_D F(\rho, \phi, \theta) \, dV = \int_{\alpha}^{\beta} \int_{g_1(\theta)}^{g_2(\theta)} \int_{f_1(\phi, \theta)}^{f_2(\phi, \theta)} F(\rho, \phi, \theta) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

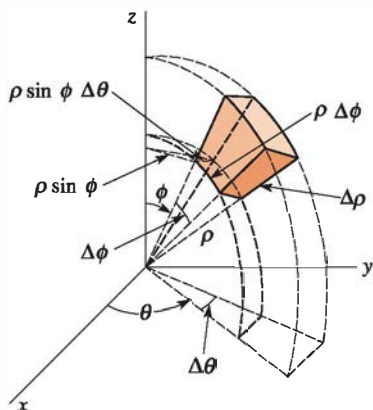


FIGURE 9.15.11 Spherical wedge

### EXAMPLE 7 Moment of Inertia

Find the moment of inertia about the  $z$ -axis of the homogeneous solid bounded between the spheres

$$x^2 + y^2 + z^2 = a^2 \quad \text{and} \quad x^2 + y^2 + z^2 = b^2, \quad a < b.$$

**SOLUTION** If  $\delta(\rho, \phi, \theta) = k$  is the density,\* then

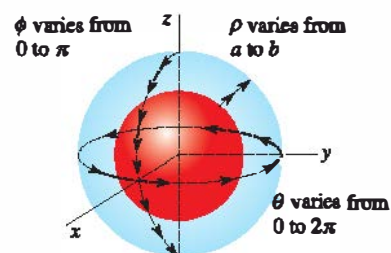
$$I_z = \iiint_D (x^2 + y^2) k \, dV.$$

\*We must use a different symbol to denote density to avoid confusion with the symbol  $\rho$  of spherical coordinates.



From (5) we find  $x^2 + y^2 = \rho^2 \sin^2 \phi$  and  $x^2 + y^2 + z^2 = \rho^2$ . Thus the equations of the spheres are simply  $\rho = a$  and  $\rho = b$ . See **FIGURE 9.15.12**. Consequently, in spherical coordinates the foregoing integral becomes

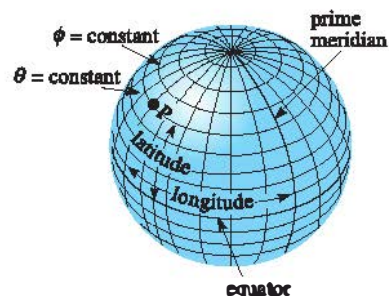
$$\begin{aligned} I_z &= k \int_0^{2\pi} \int_0^\pi \int_a^b \rho^2 \sin^2 \phi (\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta) \\ &= k \int_0^{2\pi} \int_0^\pi \int_a^b \rho^4 \sin^3 \phi \, d\rho \, d\phi \, d\theta \\ &= k \int_0^{2\pi} \int_0^\pi \left[ \frac{\rho^5}{5} \sin^3 \phi \right]_a^b d\phi \, d\theta \\ &= \frac{k}{5} (b^5 - a^5) \int_0^{2\pi} \int_0^\pi (1 - \cos^2 \phi) \sin \phi \, d\phi \, d\theta \\ &= \frac{k}{5} (b^5 - a^5) \int_0^{2\pi} \left[ -\cos \phi + \frac{1}{3} \cos^3 \phi \right]_0^\pi d\theta \\ &= \frac{4k}{15} (b^5 - a^5) \int_0^{2\pi} d\theta = \frac{8\pi k}{15} (b^5 - a^5). \end{aligned}$$



**FIGURE 9.15.12** Limits of integration in Example 7

## Remarks

Spherical coordinates are used in navigation. If we think of the Earth as a sphere of fixed radius centered at the origin, then a point  $P$  can be located by specifying two angles  $\theta$  and  $\phi$ . As shown in **FIGURE 9.15.13**, when  $\phi$  is held constant, the resulting curve is called a **parallel**. Fixed values of  $\theta$  result in curves called **great circles**. Half of one of these great circles joining the north and south poles is called a **meridian**. The intersection of a parallel and a meridian gives the position of a point  $P$ . If  $0^\circ \leq \phi \leq 180^\circ$  and  $-180^\circ \leq \theta \leq 180^\circ$ , the angles  $90^\circ - \phi$  and  $\theta$  are said to be the **latitude** and **longitude** of  $P$ , respectively. The **prime meridian** corresponds to a longitude of  $0^\circ$ . The latitude of the equator is  $0^\circ$ ; the latitudes of the north and south poles are, in turn,  $+90^\circ$  (or  $90^\circ$  North) and  $-90^\circ$  (or  $90^\circ$  South).



**FIGURE 9.15.13** Parallels and great circles

## 9.15 Exercises

Answers to selected odd-numbered problems begin on page ANS-24.

In Problems 1–8, evaluate the given iterated integral.

1.  $\int_2^4 \int_{-2}^2 \int_{-1}^1 (x + y + z) \, dx \, dy \, dz$

2.  $\int_1^3 \int_1^x \int_2^{xy} 24xy \, dz \, dy \, dx$

3.  $\int_0^6 \int_0^{6-x} \int_0^{6-x-z} dy \, dz \, dx$

4.  $\int_0^1 \int_0^{1-x} \int_0^{\sqrt{y}} 4x^2 z^3 \, dz \, dy \, dx$

5.  $\int_0^{\pi/2} \int_0^{y^2} \int_0^y \cos\left(\frac{x}{y}\right) \, dz \, dx \, dy$

6.  $\int_0^{\sqrt{2}} \int_{\sqrt{y}}^2 \int_0^{e^x} x \, dz \, dx \, dy$

7.  $\int_0^1 \int_0^1 \int_0^{1-x^2-y^2} xye^z \, dz \, dx \, dy$

8.  $\int_0^4 \int_0^{1/2} \int_0^{x^2} \frac{1}{\sqrt{x^2 - y^2}} \, dy \, dx \, dz$

9. Evaluate  $\iiint_D z \, dV$ , where  $D$  is the region in the first octant bounded by the graphs of  $y = x$ ,  $y = x - 2$ ,  $y = 1$ ,  $y = 3$ ,  $z = 0$ , and  $z = 5$ .

10. Evaluate  $\iiint_D (x^2 + y^2) \, dV$ , where  $D$  is the region bounded by the graphs of  $y = x^2$ ,  $z = 4 - y$ , and  $z = 0$ .

In Problems 11 and 12, change the indicated order of integration to each of the other five orders.

11.  $\int_0^2 \int_0^{4-2y} \int_{x+2y}^4 F(x, y, z) \, dz \, dx \, dy$